

Section A: Vector Database

A1. Explain in your own words what a vector database is and how it's different from a traditional database like MySQL or PostgreSQL.

- A vector database is a database used to store and search vector embeddings.
- It searches the numerical representation of data in the vector database.
- These vectors store the meaning of the data or words, allowing the database to find information that is similar or related.
- It performs indexing to make searches quick.

The main difference:

- MySQL/PostgreSQL mainly store structured data such as rows and columns, but a vector database stores and searches vector embeddings.
- MySQL/PostgreSQL use exact or conditional queries such as **WHERE**, but a vector database uses indexing for quick search.
- MySQL/PostgreSQL find data based on matching values, but a vector database finds data based on meaning or similarity.

A2. Take any sentence of your choice. Describe what happens to it step by step from raw text to being stored in a vector database. Name each stage.

Sentence: “My favourite sport is cricket”

1. Raw Text

The original sentence is taken as input: “My favourite sport is cricket”.

2. Text Preprocessing

The text may be cleaned by removing unnecessary spaces, special characters, or unwanted information.

3. Chunking

The sentence is divided into smaller tokens.

Example: My | favourite | sport | is | cricket

4. Embedding Generation

An embedding model converts the sentence into numerical values that represent its meaning. For example:

My	favourite	sport	is	cricket
0.24	-0.51	0.73	0.16	0.44

5. Storage in Vector Database

The generated vector is stored in a vector database, such as Pinecone, Milvus, or FAISS, along with the original text.

A3. Why can't we just use a normal database index (like a B-tree) to search through embeddings? Explain the core difference between exact lookup and similarity search.

- We use a B-tree when we want to find an exact value or a value within a fixed range.
- Embeddings are different: we do not want to find an exact vector. We want to find vectors that are similar or related to the user query.

For example: Suppose a student table has `student_id = 111`. A B-tree can quickly find `student_id = 111`, but a vector database finds the documents that are similar or related to our query.

- **Exact lookup:** used to find an exact value or values within a range.
- **Similarity search:** finds vectors that are similar to the query vector.

A4. You're given two vectors $A = [1, 2, 3]$, $B = [2, 4, 6]$. Calculate the cosine similarity between A and B by hand, showing your steps. What does the result tell you about these two vectors?

A5. Explain why a vector database is essential for RAG specifically. What would break if you removed it from the pipeline?

A vector database is an important part of RAG. It is used to store and retrieve information that is similar to the user query.

How it works in RAG:

1. Large documents are divided into smaller chunks.
2. Each chunk is converted into a numerical vector using an embedding model.
3. The vectors, along with the original text and metadata, are stored in the vector database.
4. When the user asks a question, the question is converted into a vector.
5. The vector database finds the stored document chunks that are most similar to the question.
6. The relevant chunks are given to the LLM as context. The LLM then generates an answer based on that information.

What happens if we remove the vector database: the retrieval part of RAG becomes much harder and less efficient. The system would not be able to perform fast similarity searches across many document embeddings.